Experiment 5

Aim

Create SQL views for department salary summary and employee hierarchy. Test updatability of views. Implement a recursive CTE to display reporting chains.

Source Code

1. Create Views

View 1: Department salary summary

```
CREATE VIEW v_dept_salary_summary AS
SELECT d.dept_id,
    d.dept_name,
    COUNT(e.emp_id)        AS emp_count,
    SUM(e.salary)          AS total_salary,
    ROUND(AVG(e.salary),2)  AS avg_salary,
    MIN(e.salary)          AS min_salary,
    MAX(e.salary)          AS max_salary
FROM Department d
LEFT JOIN Employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;
```

```
sql
-- View 2: Employee hierarchy (direct manager only)
CREATE VIEW v_emp_hierarchy AS
SELECT e.emp_id,
    e.emp_name,
    e.dept_id,
    e.manager_id,
    m.emp_name AS manager_name
FROM Employee e
LEFT JOIN Employee m ON e.manager_id = m.emp_id;
```

2. Test Updatability of Views

Attempt to update an aggregate view

```
UPDATE v_dept_salary_summary
SET total_salary = 1000000
WHERE dept_id = 1;
-- Expected error: ERROR 1288 (HY000): The target table v_dept_salary_summary of the UPDATE is not updatable


-- Attempt to insert into a join view
INSERT INTO v_emp_hierarchy (emp_id, emp_name, dept_id)
VALUES (99, 'Test', 1);
-- Expected error: ERROR 1394 (HY000): Can not insert into join view 'v_emp_hierarchy' without fields list
```

3. Recursive CTE for Reporting Chains

```
WITH RECURSIVE emp_chain AS (
  -- Anchor: top-level managers (no manager)
  SELECT emp_id,
      emp_name,
      manager_id,
      1 AS level,
      CAST(emp_name AS CHAR(200)) AS chain
  FROM Employee
  WHERE manager_id IS NULL

  UNION ALL

  -- Recursive: employees reporting to someone already in the chain
  SELECT e.emp_id,
      e.emp_name,
      e.manager_id,
      ec.level + 1,
      CONCAT(ec.chain, ' -> ', e.emp_name)
```

```
  FROM Employee e
  JOIN emp_chain ec ON e.manager_id = ec.emp_id
)
SELECT * FROM emp_chain
ORDER BY chain;
```

Output

Output:

```
+---------+------------+-----------+--------------+------------+------------+------------+
| dept_id | dept_name  | emp_count | total_salary | avg_salary | min_salary | max_salary |
+---------+------------+-----------+--------------+------------+------------+------------+
|       1 | HR         |         6 |    331000.00 |   55166.67 |   41000.00 |   90000.00 |
|       2 | Finance    |         6 |    361000.00 |   60166.67 |   44000.00 |   95000.00 |
|       3 | IT         |         6 |    481000.00 |   80166.67 |   60000.00 |  120000.00 |
|       4 | Sales      |         6 |    341000.00 |   56833.33 |   43000.00 |   85000.00 |
|       5 | Operations |         6 |    308000.00 |   51333.33 |   38000.00 |   78000.00 |
+---------+------------+-----------+--------------+------------+------------+------------+
+--------+----------+---------+------------+--------------+
| emp_id | emp_name | dept_id | manager_id | manager_name |
+--------+----------+---------+------------+--------------+
|      1 | Aarav    |       1 |       NULL | NULL         |
|      2 | Bhavna   |       1 |          1 | Aarav        |
|      3 | Chirag   |       1 |          1 | Aarav        |
|      4 | Divya    |       1 |          1 | Aarav        |
|      5 | Esha     |       1 |          1 | Aarav        |
|      6 | Farhan   |       1 |          1 | Aarav        |
|      7 | Gaurav   |       2 |       NULL | NULL         |
|      8 | Hina     |       2 |          7 | Gaurav       |
|      9 | Irfan    |       2 |          7 | Gaurav       |
|     10 | Jaya     |       2 |          7 | Gaurav       |
+--------+----------+---------+------------+--------------+
ERROR 1288 (HY000) at line 140: The target table v_dept_salary_summary of the UPDATE is not updatable
```